Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000;        % High-pass cutoff frequency (Hz)
Fs = 8000;        % Sampling frequency (Hz)
N = 50;           % Filter order

% Normalized cutoff frequency
wc = fp / (Fs/2);

%% --- 1. FIR High-Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));

disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect');

%% --- 2. FIR High-Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');
```

```
%% --- 3. FIR High-Pass using Hann Window ---
b_hann = fir1(N, wc, 'high', hann(N+1));

disp('--- High-Pass Coefficients (Hann Window) ---');
disp(b_hann');

%% --- Frequency Response: Rectangular Window ---
figure;
freqz(b_rect, 1, 1024, Fs);
title('FIR High-Pass Filter using Rectangular Window');
grid on;

%% --- Frequency Response: Hamming Window ---
figure;
freqz(b_hamm, 1, 1024, Fs);
title('FIR High-Pass Filter using Hamming Window');
grid on;

%% --- Frequency Response: Hann Window ---
figure;
freqz(b_hann, 1, 1024, Fs);
title('FIR High-Pass Filter using Hann Window');
grid on;
```

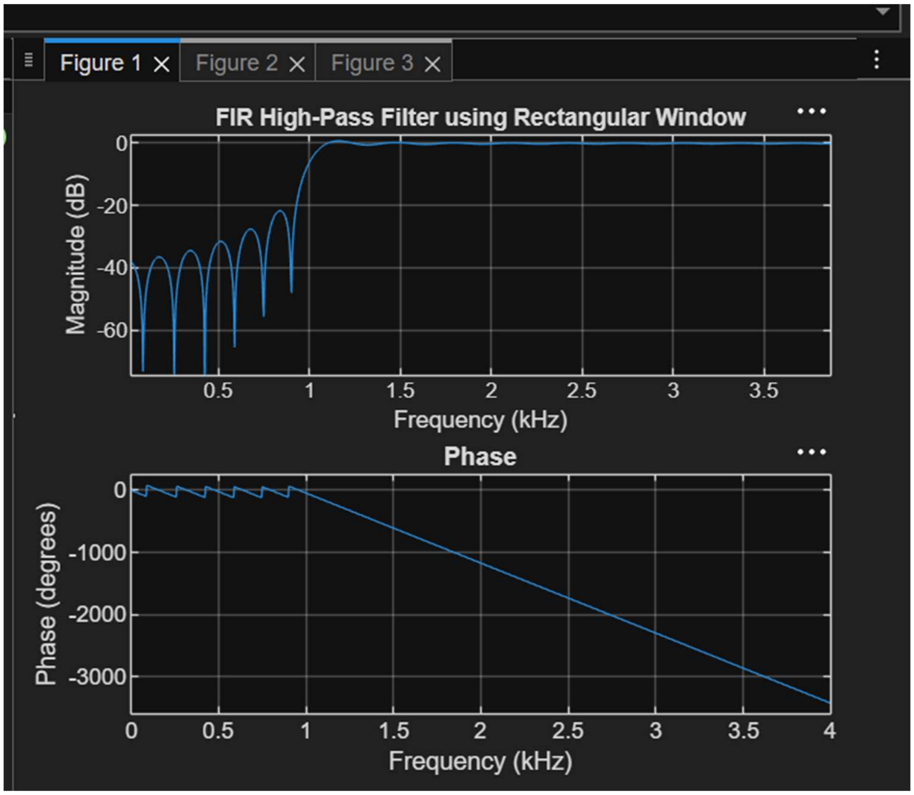


Figure 1
Figure 2
Figure 3
FIR High-Pass Filter using Rectangular Window
Magnitude (dB)
Frequency (kHz)
Phase
Phase (degrees)
Frequency (kHz)

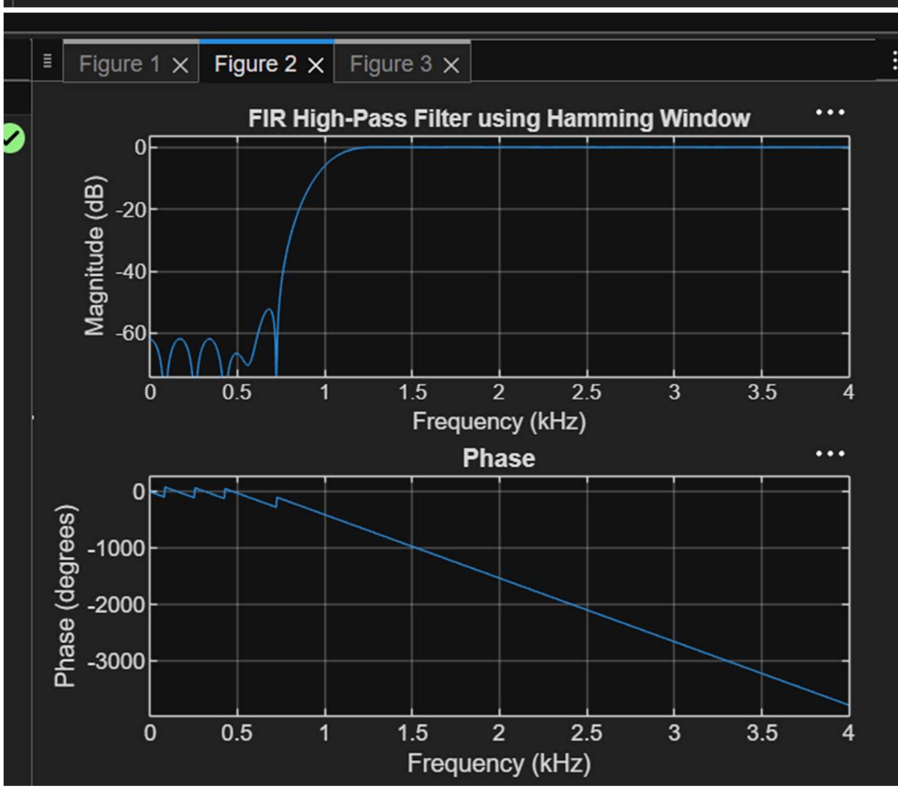
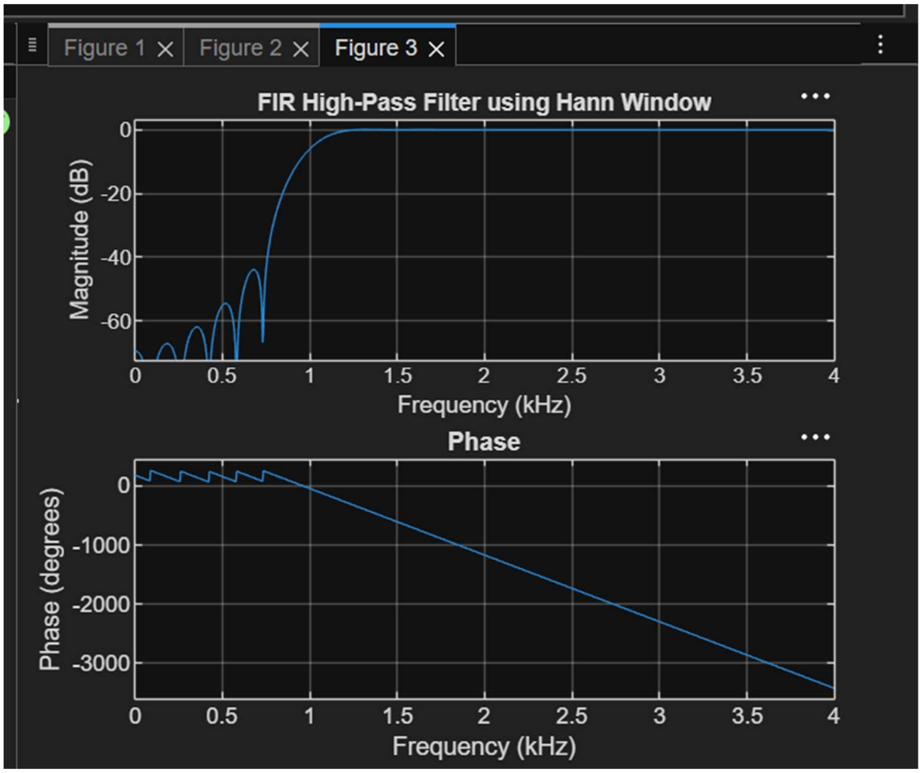


Figure 1
Figure 2
Figure 3
FIR High-Pass Filter using Hamming Window
Magnitude (dB)
Frequency (kHz)
Phase
Phase (degrees)
Frequency (kHz)



Result:

1. A discrete-time sinusoidal signal was successfully generated in MATLAB.

2. Uniform quantization was performed for different quantization levels, and the corresponding quantized signals were obtained.

3. Comparison of the original and quantized signals showed that the quantized signal approximates the original signal more closely as the number of quantization levels increases.
4. The quantization error was analyzed and found to decrease with increasing quantization levels, resulting in improved signal quality and reduced quantization noise.